Data Leakage Prevention via Vector Search for Third-Party LLM Use *

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Abstract

Enterprise reliance on third-party Large Language Model (LLM) inference providers has opened a new channel for confidential data leakage. Legacy Data Loss Prevention (DLP) tools based on byte-level fuzzy hashing and lexical matching collapse against paraphrase, translation, and summarization, where surface similarity to the source approaches zero. We propose **VS-DLP**, a learning-free semantic DLP framework using Sentence-Transformer embeddings, FAISS-HNSW indexing, and per-chunk top-1 voting. Across four datasets covering natural revisions and synthetic paraphrases, VS-DLP attains $F_1 = 0.894$ and PR-AUC $= 0.933$ at 45 ms per query on a single workstation.

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